

High-Performance Reading Machines

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Invited Paper

Reading is the process of mapping an image containing printed or handwritten text into a meaningful symbolic representation. Primary subtasks in reading are extraction of textual regions of interest, recognition of symbols (characters, words, etc.), and the interpretation of the recognized symbols using contextual knowledge. This paper describes the architecture of a reading machine designed to achieve a high rate of correct interpretation of text, as well as high speed in performing the interpretation. The refinement of the architecture for a specialized reading machine, to find and interpret addresses on a stream of postal letters, is described. The addresses can be either machine-printed or handwritten. Here, the primary subtasks correspond to finding the block of text corresponding to the destination address, recognizing characters and words within the address, and interpreting the text using postal directories. Several principles emerge from the high-performance design. They include the need for multiple algorithms for recognition, the use of multiple scales in recognition (holistic and analytic), methods for combining results of multiple algorithms, the efficacy of artificial neural nets and fuzzy matching, and the feasibility of reading unconstrained handwritten words when there exist accompanying numeric fields that limit word choices.

Keywords—Reading machines, optical character recognition, handwriting recognition, word recognition, multiclassifier systems.

I. INTRODUCTION

The development of machines that can read printed and handwritten text, with a performance level similar to that of humans, has long been a goal with economic and scientific importance.

Its economic importance arises from numerous applications, among them automatic sorting of postal mail pieces, reading machines for the blind, data entry from bank checks, and processing manually filled-out tax forms.

The scientific challenge arises from the artificial intelligence goal of developing machines to perform tasks that when done by humans are said to require intelligence. As a human skill, reading is learned. It has long been a topic of study by cognitive and educational psychologists, whose studies can be useful for those interested in de-

veloping reading machines. The reading of text on paper encompasses both perception and language (and is thus said to involve both the left and right parts of the brain). Both perception and language have long been frontiers for artificial intelligence research. Reading poses the challenge of naturally integrating methods from both domains.

There are many goals in human reading. Some examples are reading to relieve boredom, searching for a telephone number in a directory, reading an address on an envelope to determine its destination, reading poetry, reading as an editor (primarily concerned with style and punctuation, not content), reading a newspaper article, and reading a textbook for content. According to Singer [1], "In reading along a particular sentence, a reader must retrieve and mobilize systems for recognizing words and phrases, next link the recognized words or phrases to their corresponding meanings, and subsequently utilize various cognitive processes for inferring, interpreting, and inductively, or deductively arriving at conclusions or solutions to problems."

For the purpose of developing reading machines, we will define the process of reading as the process of mapping a visual field containing iconic representations of text (handwriting or machine printing) into an appropriate symbolic representation. A high-performance system performs this task with a high degree of accuracy and at high speed.

This paper describes a general architecture for a reading machine and its specialization to the task of determining the destination of each letter in a letter stream.

II. AN ARCHITECTURE FOR READING MACHINES

The major steps in reading text in a given visual field are image acquisition, layout analysis, and reading the text (Fig. 1).

A. Image Acquisition

Image acquisition is the process of scanning a document so as to represent its image as a gray-scale array. Four basic building blocks form functional image scanners: a detector (and associated electronics), an illumination source, a scan lens, and a document transport [2]. To capture an image, the object is placed in the scanning field, a light source floods the object with illumination, and the lens forms an

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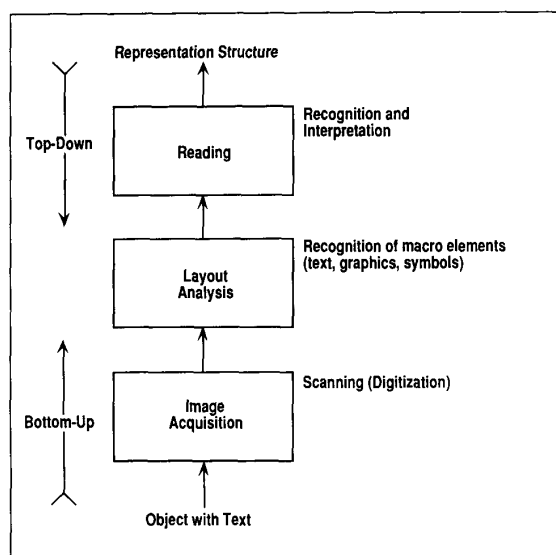


Fig. 1. Reading text is a bottom-up and top-down process.

image on the detector. The detector consists of an array of elements each of which converts incident light into a charge, or analog signal. These analog signals are then converted into a digital image. Scanning is performed by the detector and the motion of the text object with respect to the detector.

Three types of integrated circuits dominate the scanner market today: photodiode arrays, charge-coupled device (CCD) arrays, and charge-coupled photodiode device (CCPD arrays). They differ in how light is converted into analog signals and how signals are transferred off chip. Time delay integration (TDI) CCD's operate under low illumination levels since they collect photons over a larger temporal space. TDI arrays are also preferable for high-speed scanning applications.

For a fixed quantization level (number of gray levels), image resolution needed to perform layout analysis is much lower than that for recognizing characters and words. The image acquisition module is expected to scan and digitize the image at a single resolution, however. A resolution of 300 pixels per inch (ppi), 8 bit gray scale, is sufficient for recognizing 8-point printed text and thin strokes in handwriting. A down-sampled image (around 100 ppi) is input to the layout analysis subsystem, which latter determines the spatial extent of the higher resolution subimage to be presented to document recognition.

Since text is printed as dark points on light backgrounds (or vice versa) binary images are generally considered to be sufficient for both layout analysis and document reading. However, gray-scale images can enhance recognition rate, particularly for poor quality images, at an increase in processing expense. The scanned gray-scale image can be converted to binary using thresholding algorithms. For documents with high contrast, such as those with printing or handwriting on a white background, a global threshold

obtained from a gray-scale histogram is sufficient [3]. For other documents it may be necessary to adapt to local contrast [4].

B. Layout Analysis

The task of locating textual regions of interest is the primary function of layout analysis. The task of reading involves recognizing shapes (of characters, words, etc.), i.e., mapping icons to symbols, and interpreting the symbols, i.e., mapping low-level symbols to higher-level symbols. The location task is often separable from the recognition/reading task; for example, it is possible to tell where the text is present even if it is unreadable. However, a system interactive with recognition is needed for nonstandard text structures such as handwritten documents.

Layout analysis involves both segmentation of the document into textual, graphical, and other regions and labeling certain textual regions according to their logical role in the document. Recognition is performed on the resulting components so as to hypothesize the logical role of textual regions.

C. Reading and Understanding

The task of recognizing textual symbols and interpreting the symbols using knowledge is referred to as reading. When the reading task involves the interpretation of text as well as accompanying photographs and graphics (e.g. drawings and graphs), the task may be more appropriately referred to as document understanding [5].

An architecture for reading/document understanding is given in Fig. 2. This is a refinement of the process illustrated in Fig. 1. The flow of data is from the point of view of a high-speed implementation. It shows a recognition-based approach to layout analysis. Preliminary recognition of connected components allows hypothesization of the nature of the document region. Final assignment of region labels is based on a more global analysis.

A complex document, e.g., a page of a daily newspaper, consists of text, photographs, and graphics. The understanding task is to begin with the image of a page, extract a story of interest, and map the textual and accompanying graphical information into a symbolic representation (such as a semantic network).

For a specialization of the architecture to newspaper images, the reader is referred to [6]. The text location task for newspaper images, which can exploit the regular layout usually employed, is discussed in [7]. The reading of degraded text encountered in newsprint is discussed in [8], and the integration of information from textual and photographic information in newspapers is the topic of [9].

III. MAIL PIECE UNDERSTANDING

Reading machines play an important economic role in the processing of documents. Document classes commonly encountered include business letters, journals, books, newspapers, addressed envelopes, and forms. It is useful to

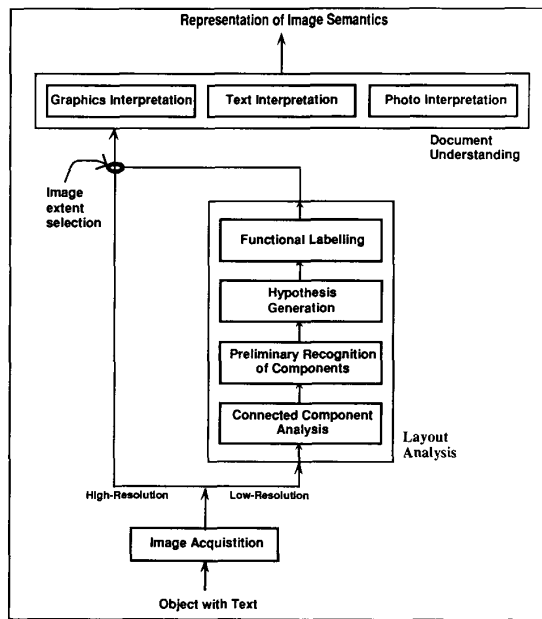


Fig. 2. Architecture for a general reading machine.

examine how the architecture of Fig. 2 specializes to document classes.

An important application of reading systems is in the domain of postal mail sorting. This is a goal-oriented reading task, where the objective is to determine the destination of a piece of mail from an image of its face. Mail pieces are usually divided into different classes, such as letters (e.g., envelopes and postcards), flats (e.g., magazines), and parcels. Postal address reading differs from most other document reading tasks owing to the wide variability of fonts and printing qualities encountered, the presence of either machine printing or handwriting, the variability of document layout, etc.

There exists a natural representation for the semantics of an address within the United States. The task of the U.S. Postal Service address recognition unit is to encode each letter mail piece by a string of digits, the first five of which are referred to as the ZIP code, together with the next four as the ZIP + 4 code, and together with two more as the 11-digit destination code (ZIP + 4 plus an additional two digits corresponding to the last two digits of the street address). Once the destination encoding has been determined, it is then sprayed as a bar code on the face of the mail piece.

The subsystems specialize to: (i) finding a block of text corresponding to the destination address, and (ii) reading handwritten and machine-printed postal addresses (Fig. 3). The tasks of interpreting machine-printed addresses and handwritten addresses are divided into different streams, because the techniques involve several differences: handwriting consists of elongated strokes whereas machine print consists of regular blacks; the shapes of characters in handwritten words are dependent on word context;

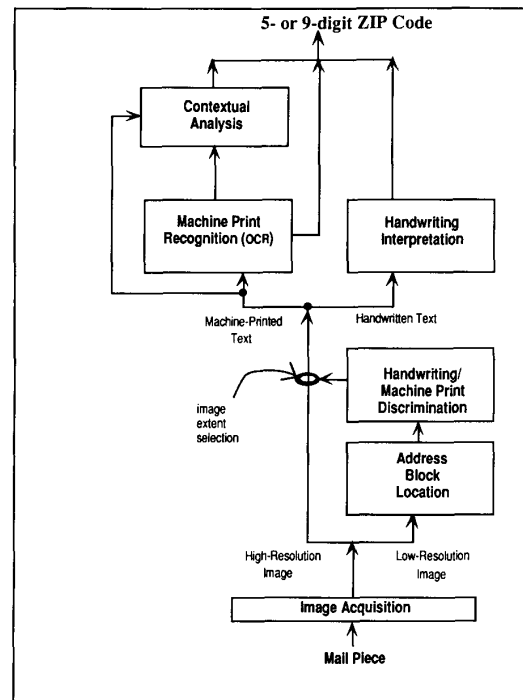


Fig. 3. Address recognition unit system architecture.

handwriting reading requires greater contextual interaction whereas machine print reading can be separated into character recognition and contextual analysis; etc.

The need to process large numbers of letters in short time spans translates into a stringent requirement on the throughput. The address recognition unit needs to process 13 letters per second. The economics of postal mail sorting requires that 90% of machine-printed and 50% of handwritten addresses be destination encoded and that the error rate be kept low (roughly 0.5% overall).

Organization of Rest of Paper

The remainder of this paper focuses on a high-performance system, under development in our laboratory, to find and interpret destination addresses on postal mail pieces. Section IV describes a methodology and implementation for the task of finding a text block of interest, i.e., the destination address.

The process of interpreting a text block consisting of a handwritten United States address is described in Section V. It involves the primary task of locating the ZIP code and segmenting and recognizing its digits and the secondary task of recognizing handwritten words (city and street names) by using all contextual information available at that point.

The process of interpreting machine-printed text using a large data base of knowledge is considered in Section VI. The focus is on recognizing poor-quality machine-printed characters and the analysis of entire addresses using postal directories.

IV. GOAL-ORIENTED LAYOUT ANALYSIS: ADDRESS BLOCK LOCATION

In goal-oriented document layout analysis, the document is analyzed only to the extent that it is necessary to find one or more fields of interest. Address block location (ABL) is such a goal-oriented task where the input is an image of the face of an arbitrary mail piece. The objective is to extract one or more rectangular subregions of the image as candidates for the destination address block (DAB).

A typical mail piece has several regions or blocks that are meaningful to mail processing, e.g., address blocks (destination and return) and postage (meter mark or stamp), as well as extraneous blocks such as advertising text, logos, and graphics. The significant blocks have certain characteristic properties and spatial relationships that can be derived from images and used to assist in the task.

The task is one of finding regions of interest in either a structured or a cluttered environment. The regions of interest are blocks of machine-printed or handwritten text. The complexity of the task depends on whether the mail piece is a letter (which usually has structure), is flat (where the address block may be randomly placed), or is a parcel. The handwriting/machine print discrimination function is a by-product of this operation.

The approach taken was to first develop a set of tools to segment the image and to detect characteristic properties that are relevant to assigning appropriate labels to the segmented regions. The design was arrived at by examining a large data base of characteristics of mail pieces relevant to ABL. The design uses several specialized functional tools [10]. The tools are divided into two categories: image analysis tools and block analysis tools (see Table 1). The basic system consists of three image analysis tools and 11 block analysis tools. The image analysis tools take images as input and produce other images or symbolic information, e.g. gray-scale to binary and extraction of extent of subimages. The block analysis tools take symbolic description as input, e.g. extents of different blocks and confidences associated with blocks for different labels, and produce other symbolic descriptions.

Segmentation is the process of generating candidate regions by analyzing the binary image. Due to the fact that addresses have several different visual structures, several segmentation tools are used. One specializes in handwriting (which consists of elongated thin strokes) and another specializes in machine print (which consists of regularly spaced blobs). Segmentation is performed by labeling all connected components and then determining a region-adjacency graph. Components having similar sizes and near each other are connected together in this graph. Clusters of connected regions in this graph represent blocks of text. Different size parameters are used for handwriting and machine-print.

A. Real-Time ABLS

The goal of real-time ABLS (RT-ABLS) is to achieve a throughput of 13 letters per second. The architecture of RT-ABLS involves both pipelining and parallelism (Fig. 4)

Table 1 Functional Descriptions of Tools in basic ABL System

Category	Tool	Description
Image Analysis Tools	DIGI	Produces digitized images of the original document
	ADTH	Adaptive thresholding to convert a gray-scale image into a binary image using local contrast.
	MSEG	Segmenter to locate connected components corresponding to machine-print and group them into blocks using a region adjacency graphy. Consists of MSEG1 and MSEG2.
	HSEG	Segmenter to locate handwritten blocks by first filtering blobs using HSEGF and then using HSEG1 and HSEG2 (analogous to MSEG1 and MSEG2).
Block Analysis Tools	LSEG	A segmenter that groups blobs into lines and lines into blocks. Uses output of MSEG1.
	ZIPM	Merges ZIP code block in lower right of a destination address candidate.
	BLCS	Splits a too high or too wide machine-generated text block into several smaller text blocks.
	SIZE	Uses block features, e.g., aspect ratio, length, height, number of text lines, and number of components, etc., to classify how likely a block is a destination address, return address, or advertising text.
	LAYO	Examines the layout of text lines in a block.
	LOCA	Uses the location of a block to determine the likelihood of this block being the destination address, return address, or postage.
	HEUR	Uses spatial heuristics or rule-of-thumb to guess the destination address from a list of candidates.
	COVF	Verifies the consistency of labeling hypotheses among neighboring blocks.
	UNIF	Unifies the block features between blocks generated by different tools.
	EVHP	Pools together the evidence generated by various tools and generates labeling hypotheses.
	STOP	Decides whether to halt processing or not.

[11]. The pipeline design involves four image processing tools: ADTH (adaptive thresholding), MSEG (machine-print segmentation), HSEG (handwriting segmentation), and LSEG (line segmentation). This design feeds the result of ADTH to MSEG, HSEG, and LSEG in parallel. The HSEG operation is itself broken into three stages: HSEGF, HSEG1, and HSEG2. The MSEG operation is divided into two stages: MSEG1 and MSEG2.

The first operational step within HSEG, viz. HSEGF, converts a handwritten image into vertical strokes by deleting certain foreground pixels, predominantly ligatures. Both HSEG1 and MSEG1 perform essentially the same task of connected component labeling and grouping but slightly different parameters for eliminating noise components. Similarly, MSEG2 and HSEG2 are identical but involve group-

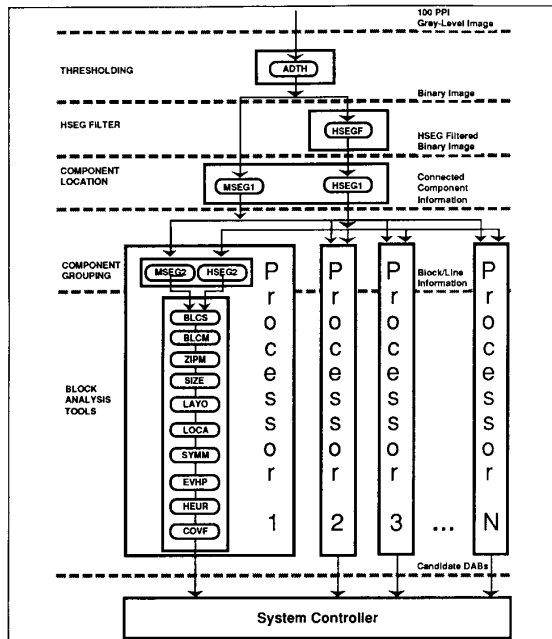


Fig. 4. Real-time address block location architecture.

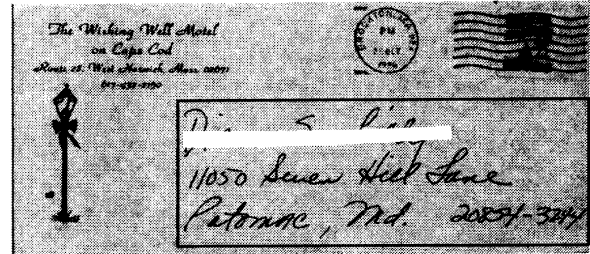
ing components into blocks. Special-purpose hardware, which employs pipeline processing of pixels (based on the Datacube MAXBUS standard), is used to perform the ADTH, HSEG, and MSEG functions. Such a pipeline processes pixels at the rate of 100 ns per pixel. Based on the speed requirement, a resolution of 100 ppi was determined to be sufficient.

B. Performance

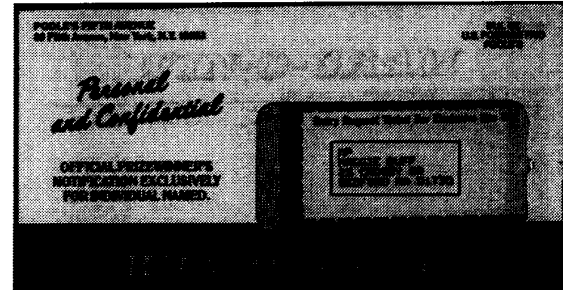
Examples of images with handwritten and machine-printed address blocks located by RT-ABLS are shown in Fig. 5. RT-ABLS was tested on 2046 MLOC (multiline postal OCR) binary images, all from the letter mail class. The images were not destination encoded by the postal OCR. They were processed in the form of down-averaged 100 ppi binary images. The resulting performance was as follows: success (correct top choice) = 88.3%, partial success (city, state, ZIP included) = 1%, correct reject (top candidate was not DAB) = 0.4%, wrong reject (top candidate was DAB) = 1.3%, type 1 error (something other than return address chosen) = 7.7%, and type 2 error (return address chosen) = 1.2%.

C. Window Detection

The performance of RT-ABLS can be aided by a hardware system that can detect the presence of windows on envelopes. Glassine windows on envelopes can be detected from the reflective properties of the material. By placing sensor pairs at a critical angle, the amount of reflectance can be measured. The output of the sensors is processed by a high-speed digital signal processor (DSP). This processor



(a)



(b)

Fig. 5. Examples of address block location performance: (a) image with handwritten address, where the located address block is shown with bounding box; (b) image with machine-printed address and located address block.

calculates the coordinates of the window that is present on the mail piece. The result of window detection can be used as additional segmentation.

D. Handwriting/Machine-Print Discrimination

Discrimination between handwriting and machine print can be performed using symbolic information. Connected components generated by the address block location process can be used to determine whether the candidate block generated has the characteristics of handwriting or machine printing. Three properties of blocks are useful for performing the discrimination: standard deviation of blob heights, standard deviation of blob widths, and average component density. Furthermore, a single feature consisting of the number of different blob heights provides a high level of discrimination.

The basic idea is that machine-printed text is much more uniform than handwritten text. Hence the heights of connected components within a block will be much more uniform for machine print, e.g. the heights of capital letters will be of the same height. The number of different blob heights, n , will be much larger for handwritten text than for machine-printed text. So a threshold value can be established such that if n exceeds the threshold, the block is handwritten; otherwise it is machine printed.

V. HANDWRITTEN TEXT INTERPRETATION

Interpretation of handwritten addresses is an example of the task of reading handwriting in a structured environment.

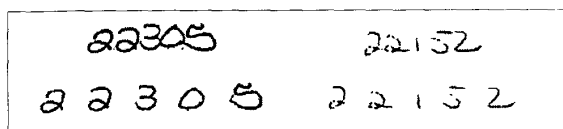


Fig. 6. Successful segmentations of difficult cases.

Other such domains are bank checks and drug prescriptions. The interpretation involves locating the text of main interest, viz. the ZIP code if present, then reading the ZIP code and resolving ambiguity using other text fields. There exists a large literature on the recognition of handwritten characters, digits, and words; however, most pertains to the on-line case where writing is on an electronic tablet [12]. The off-line case, the one of interest in document recognition, is more difficult. Cursive writing and mixtures of discrete and cursive segments are particularly difficult, unless there exists a lexicon of possible interpretations.

The recognition of digit strings is usually easier than alphabetic words in a given document. There are fewer digits than alphabets and digits touch each other less often. Thus a strategy for reading a block of handwritten text is to first recognize digit strings and then exploit constraints provided by digit strings to hypothesize the identity of alphabetic words.

Our discussion of the handwritten text interpretation problem is divided into digit string segmentation and recognition, word recognition, and address interpretation.

A. Digit String Segmentation

Given the image of a string of digits, the goal of the segmentation algorithm is to partition the image into regions, each containing an isolated digit. The technique consists of the following major phases: connected component analysis, estimation of the number of digits (obtained by involving a digit recognizer), and grouping and dissection. Examples of the successful segmentation of difficult cases involving touching characters and broken characters are shown in Fig. 6.

B. Digit Recognition

Handwritten digits have wide variability—from neat printing to broken and touching characters. Experiments indicate that for robust digit recognition it is advantageous to use multiple approaches and combine the results. The power of widely varying methodologies (statistical, structural, rule-based, neural nets, etc.) can thus be exploited. Digit recognition is performed using a suite of independent and uncorrelated digit recognition algorithms whose decisions are merged using a combination scheme [13] (Fig. 7).

Several approaches to combining decisions from multiple experts are possible. The principle of using multiple algorithms to compensate for mutual deficiencies in commercial OCR has been previously suggested [14]. However, optimality is difficult to achieve owing to practical limitations such as finite data, high dimensionality, and unknown

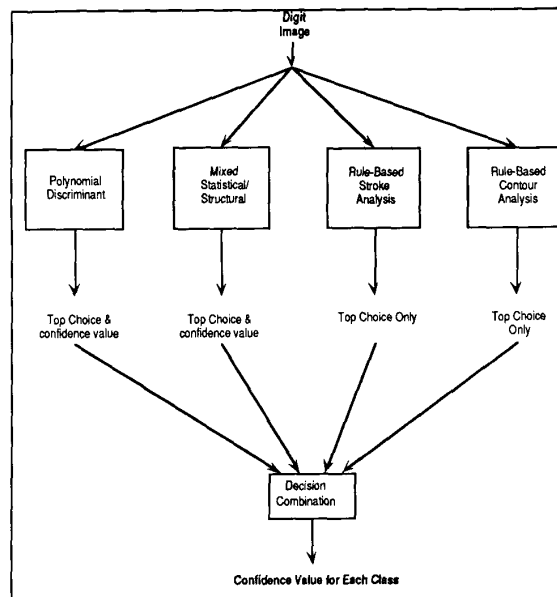


Fig. 7. Digit recognition using multiple classifiers and a decision combiner.

density functions. Artificial neural networks have been shown to approximate Bayesian decisions [15]. Thus a neural network provides an effective means of estimating the optimal decision, without dealing with those limitations directly. Input consists of decisions from multiple classifiers, represented in the form of a vector of confidence values for all classes. Four digit recognizers were employed: polynomial discriminant function [16], a mixed approach classifier [17], a stroke-based recognizer [18], and a structural contour-based chain-code classifier [19], whose individual performances were 93.6%, 89.4%, 86%, and 84.1% on the same testing set. The neural-network decision combiner achieved 96% accuracy on handwritten digits. This accuracy surpasses results obtained by other decision combination algorithms that were based on decision trees, Dempster-Shafer evidence combination, and a Bayesian formalism.

C. Word Recognition

The word recognition procedure accepts a word image as input, as well as a lexicon, i.e., a list of possible identities of the word (which may be derived from knowing the ZIP code and a postal directory). Its goal is to provide an ordering of words in a lexicon, ordered according to the level of match with the word image. Several approaches to the word recognition problem have been or are being explored in our laboratory [12], [20], [21].

One method takes a word image and determines its connected components. Each component is classified as either a discrete character or a cursive word segment using criteria based on the width/height ratio. Discrete characters are recognized by calling a discrete character recognition

system. Cursive segments are analyzed by a partial cursive class recognition system. If the component type is ambiguous, somewhere between discrete and cursive, both systems are called and the results of each system are used. At the end of this component identification and recognition process, all results are concatenated and used with a matching algorithm to sort the dictionary according to the level of match.

A second approach to cursive word recognition overcomes some of the limitations of previous approaches, such as small dictionary size. In order to recognize words involving a large dictionary, e.g., city names in the U.S., it is necessary to recognize many more characters in the word. The approach is based on segmentation and recognition of individual cursive characters in the word. Since most handwriting contains predictable flaws, the segmentation process is actually a search that generates a series of segmentations of the word. Potential characters that result from segmentation are passed to character recognition, which ranks each segment with respect to each of the 26 characters. The system keeps track of the rank of each character as segmentation proceeds. Depending on the desired depth of search, the top n characters and their respective segmentations are chosen and the segmentation process continues from that segmentation point. This results in an n -way branch at each character candidate segmentation point in the word. The final result is a series of strings which represent the characters detected from each segmentation sequence of the word. Since many segmentations result in words that contain improbable (and illegal) sequences of characters, most strings are rejected. The remaining strings are passed to a matching algorithm for dictionary matching and ranking.

The third approach to handwritten word recognition is based on first dividing the word into segments and then recognizing the segment string. The recognition method is based on the hidden Markov model. The observed segments are referred to as symbols and the characters giving rise to the symbols are the hidden states. The Viterbi algorithm and a dictionary are used in the final recognition [22].

Ranked outputs of several uncorrelated word recognition algorithms can be combined so as to improve overall performance. One combination method, known as the Borda count method, assigns a rank to each word based on the sum of the numbers of entries below it within each ranked list [23]. Two word recognition algorithms were considered for combination. Both were tested on images of 91 handwritten street names involving a dictionary of five words. The first resulted in performances of 79% and 90% for the top one and two choices. The second had performances of 85% and 90%. The combination performances were 89% and 97%.

D. Control Structure

The basic ZIP code recognition procedure performs the task of locating the ZIP code within the address, followed by segmenting and recognizing the ZIP code digits. There are various characteristics that make the task nontrivial: the ZIP code is often on the last line (90% of cases),

but not always; underlines make segmentation of lines and digits difficult; poorly formed addresses without a ZIP code may be incorrectly identified as having a ZIP code; the segmentation of a ZIP code into digits is difficult when digits touch, have connecting ligatures, or have overhangs between digits; the recognition of digits needs to be able to handle a variety of writing styles and the artifacts arising from segmentation.

A complete system for determining ZIP codes from addresses written with unconstrained handwriting requires many components [11], [24]. Figure 8 illustrates. It may be necessary to first perform preprocessing operations such as thresholding and underline removal. The text lines have to be determined and separated into words, and candidates for the ZIP code and other keywords (city, state, P.O. box, etc.) have to be determined. The flow of control in the basic handwritten ZIP code recognition system is given below. The input is a high-resolution (300 ppi) gray-scale image of a handwritten address block. The performance of this system is 75% accept with a 1.5% error rate of those accepted.

Procedure Basic Handwritten ZIP Code Recognition

1. Input Gray-Scale Image of Address Block

2. Preprocessing

- 2.1 Remove Horizontal Lines;
- 2.2 Label Connected Components;

3. ZIP Code Location

- 3.1 Perform Line Segmentation;
- 3.2 Generate ZIP Code Candidates;
- 3.3 Order Candidates by Location Confidence;

4. ZIP Code Recognition

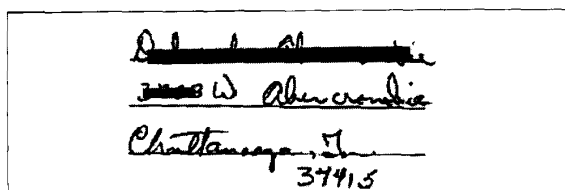
Do While a ZIP Code Candidate exists *AND* Fewer than 2 Candidates tried

- 4.1 Select Highest Ordered Candidate;
- 4.2 Segment Candidate into Digits;
- 4.3 Recognize Digits;
- 4.4 Assign Confidence (based on digit recognition) to Candidate;
- 4.5 If Confidence > Threshold *AND* Zip Code is valid *Then Return* Zip Code and Exit;
- 4.6 Remove Current Candidate from Further Consideration;

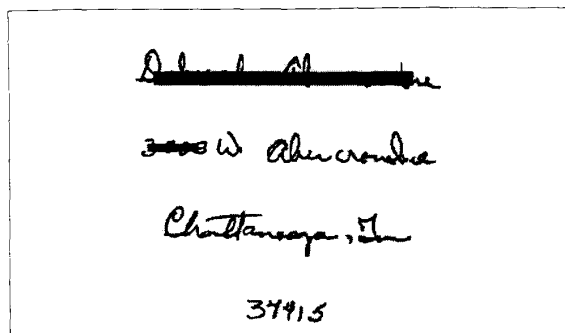
5. Reject Image

The basic handwritten ZIP code recognition procedure (HZR) does not involve reading city, state, and street words. This next level of performance improvement is to incorporate the capability to recognize handwritten characters and words into the system.

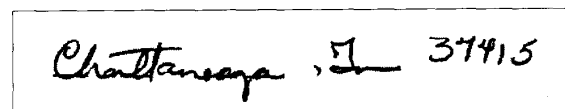
The objective of handwritten address interpretation is to integrate the handwritten ZIP code recognition procedure with the handwritten word recognition procedure as well as a postal directory lookup in order to achieve destination encoding. Because of the difficulty in correctly recognizing



(a)



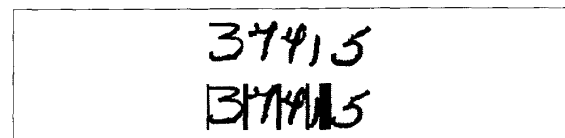
(b)



(c)

0.90	text	0.90	box	0.49	5digits
	0.90	text	0.20	text	

(d)



(e)

3	0.999000
7	0.667000
4	0.979000
1	0.999000
5	0.999000

(f)

Fig. 8 Example of handwritten ZIP Code recognition operations: (a) thresholded handwritten address image (with some fields blacked out); (b) result of finding lines of text; (c) city-state-ZIP code complex with three words identified; (d) confidence levels associated with each label for each word; (e) segmentation of ZIP code field into digits; and (f) highest-level confidence digit recognition results.

handwritten characters and words, it is necessary to use contextual information, such as a directory that relates ZIP codes with street addresses (called the ZIP + 4 directory).

11 Dec.

703

West Ferry St

Buffalo NY

14222

Ordered Lexicon of Street Names

14222-1658	703 W. FERRY ST
14222-1416	703 AUBURN AVE
14222-1220	703 W DELAVAN AVE #1
14222-1449	703 LAFAYETTE AVE

ZIP+4 Code selected

14222-1658

Fig. 9. Handwritten address interpretation.

An example is shown in Fig. 9, where recognition of the ZIP code and the street number allows the street name lexicon to contain only four names. Thus the recognition problem boils down to a problem of hypothesis testing.

VI. MACHINE-PRINTED TEXT INTERPRETATION

The goal of address interpretation is to determine the destination code. This involves reading the destination code if explicitly present in the address (e.g., a ZIP+4 code) or inferring one from the street address, city, state, etc. The goal is similar to the handwritten case. However, the volume of machine-printed addresses is much larger (85% versus 15%), thereby imposing higher performance requirements. Also, machine-printed addresses are often destined to businesses which have more complex address structures.

Our discussion of machine-printed address interpretation is divided into character recognition, word recognition, and contextual analysis.

A. Character Recognition

Machine-printed character recognition has a very large literature; there are several surveys, including ours [25]. We describe here three methods: polynomial, fuzzy template matching, and Gabor feature-based. Each method was trained on 47 555 alphanumeric images and tested on 26 357 different images. All images are 300 ppi, gray scale. One method for machine-print recognition that has found significant success in postal reading is based on polynomial discriminant functions [16]. Our implementation of the polynomial method uses certain pixel pairs from a normalized 16×16 image as features. It yields a correct recognition rate of 89% on binary images and 92% on gray-scale images.

A second method for character recognition that we have found to perform well on poor-quality binary images of machine-printed characters is based on fuzzy template matching. In character recognition, the absolute positions of pixels in an image are irrelevant to its recognition. In addition, missing pixels of an image can be partially recovered from their neighbors. Based on these ideas, we developed a fuzzy template-matching procedure for character recognition. The key idea of this fuzzy matcher is to distinguish between different degrees of mismatch. In template matching, two pixels having different values have a difference of 1. In fuzzy template matching, the difference of two pixels depends also on their neighbors. The top choice performance of the fuzzy matcher was 89% while the performance of binary template matching was 80%.

A third method of character recognition with significant promise, particularly for gray-scale recognition, is one based on Gabor functions [26]. The optimality of Gabor functions in maximizing the joint resolution in the spatial-frequency information domain was established by Gabor in 1946. Interest in their application to computer vision, pattern recognition, data compression, and other areas started only after the discovery of their resemblance to striate cortical cell profiles in 1980. A set of selected Gabor functions act as spatially localized, oriented feature extractors. Least-square solutions of those function coefficients provide compact representations of original images. Classification was accomplished by a neural network trained on those coefficients. Experiments on 300 ppi, gray-scale machine-printed characters show 98.2% accuracy on digits and 93% accuracy on correctly segmented 62 class alphanumerics.

B. Word Recognition

A typical address consists of four lines, each with four five-letter words. Thus it consists of 80 letters. Assuming a 95% correct character recognition rate yields four character errors, or only a 75% correct word recognition rate. In order to achieve a significantly higher word recognition rate, it is necessary to go beyond the character level and utilize word context.

A word recognition algorithm associates a word image with one or more entries in a lexicon of possible word identities. Knowledge about the context of characters that occur within words in the lexicon has been used to improve the matching process. For example, it is well known that the identity of a character is constrained by the identity of its predecessor within a word. There are many such constraints and a number of techniques have been developed to utilize them. Each approach has its particular strengths and is applicable to a subset of the word recognition problem.

Methods for using contextual knowledge in word recognition can be divided into three classes: character-based, segmentation-based, and word shape analysis. Character-based and segmentation-based methods are analytical methods, whereas word shape analysis is a holistic method. Each approach uses image information in recognition in a different way. Figure 10 illustrates.

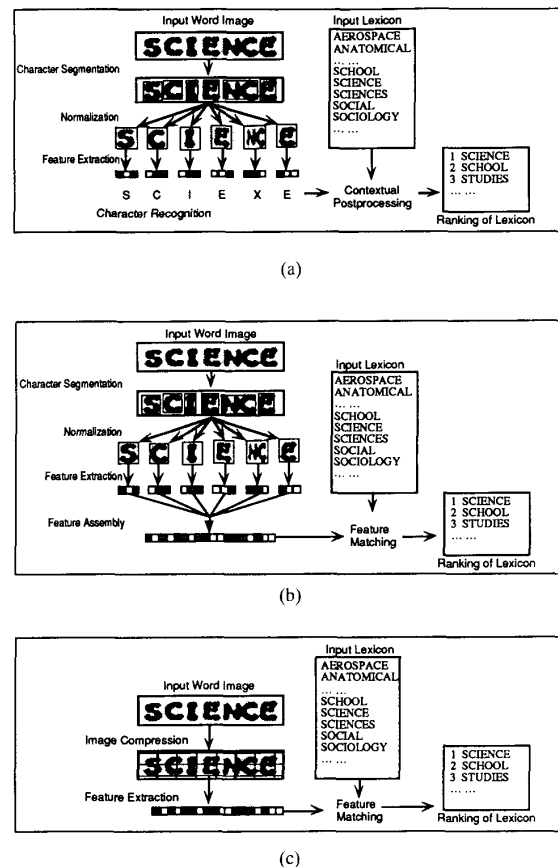


Fig. 10. Machine-printed word recognition: (a) character recognition approach; (b) segmentation-based word recognition approach; and (c) word-shape analysis approach.

In character-based word recognition, individual characters in a word are first isolated and recognized. Character decisions are then postprocessed with a dictionary of allowable words to correct character recognition errors. These methods are suitable for cases where a reliable segmentation can be obtained and the segmented characters are not deformed by normalization. It is an appropriate strategy for shorter words which are easier to segment, and where little word-level context can be utilized.

An alternative to character-based techniques is to defer decisions about character identity and to perform segmentation-based word recognition. This is suitable for images where the characters can be easily extracted but are difficult to recognize in isolation. In this approach, feature descriptions of the extracted characters are assembled and matched to a similar representation of the word in a lexicon. These techniques effectively focus word-level knowledge on the recognition process and are suitable for situations where characters can be segmented and better recognized together with other characters in the word.

A third type of word recognition algorithm determines features from the whole word shape and uses this description to calculate a group of words in a dictionary that match

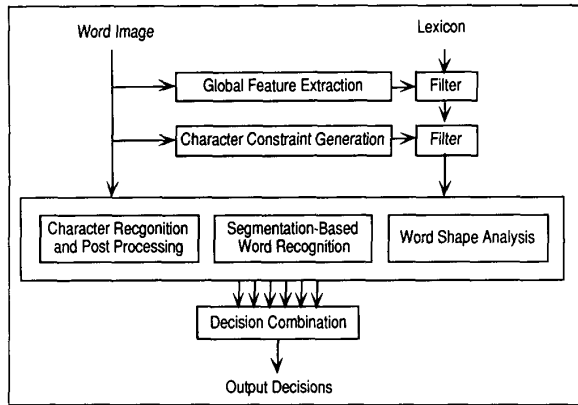


Fig. 11. Word recognition algorithm.

the input. These techniques have been further refined to yield methods that can perform high-accuracy recognition. Such methods are suitable for images that are difficult to segment into characters or when characters are distorted after extraction and normalization in isolation.

Each method uses contextual information at a different level. In the character recognition approach, contextual constraints are used only in the last stage, that is, the postprocessing stage after the character classes are decided. In the segmentation-based approach, contextual information is used before class decisions are made, but after feature extraction. In the word shape analysis approach, word context information is used directly in feature computation and matching.

A robust word recognition algorithm to integrate all these methods has been developed [23]. Reliability is achieved by making use of multiple classifiers with uncorrelated errors at various stages. Final decisions are derived by applying a group consensus function that measures agreement among parallel classifiers (Fig. 11).

The utility of this approach was demonstrated by an application to machine-printed words with a wide range of font styles and quality. The data base is a collection of images of machine-printed postal words obtained from live mail. They were scanned on a postal optical character reader at roughly 200 pixels per inch and binarized. Lexicons derived with domain knowledge are expected to contain thousands of words. Early results show that the integrated algorithm performs significantly better than each of the three approaches.

The method has been applied to a data base of postal word images scanned on an OCR at 200 pixels per inch. In many cases the image quality was very poor. A typical lexicon in the experiment contained 500 words. Applying the method to a testing set of 1710 images resulted in a 96% correct rate within the top ten choices. This method can be employed with an existing machine-print recognizer to achieve a robust word recognition system. The resulting system can be applied to recognize words in various positions in an address block image so that contextual

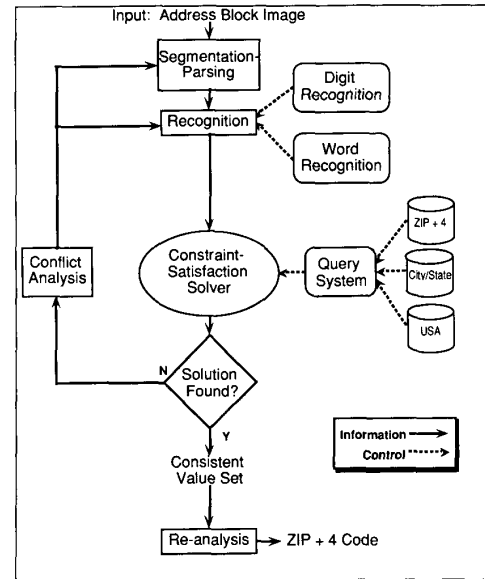


Fig. 12. Processing steps in contextual address analysis.

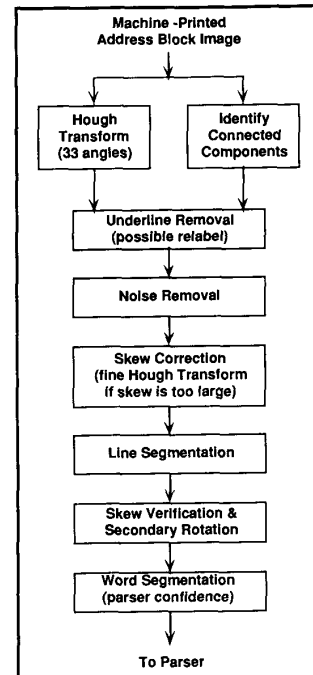
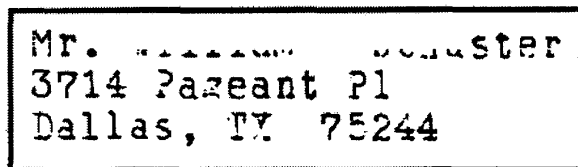


Fig. 13. Preprocessing and segmentation of machine-printed addresses.

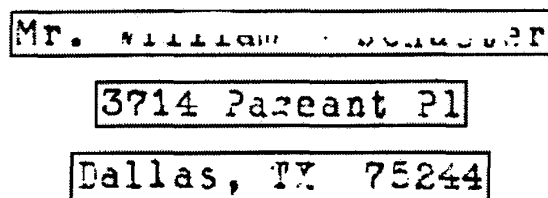
address information can be utilized to improve the currently achievable level of sort.

C. Contextual Analysis of Addresses

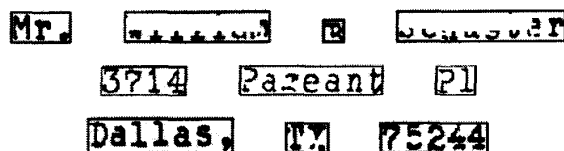
Assigning a nine-digit ZIP + 4 code to a mail piece that may only contain a five-digit ZIP code is of central importance in automated mail sorting. A nine-digit ZIP +



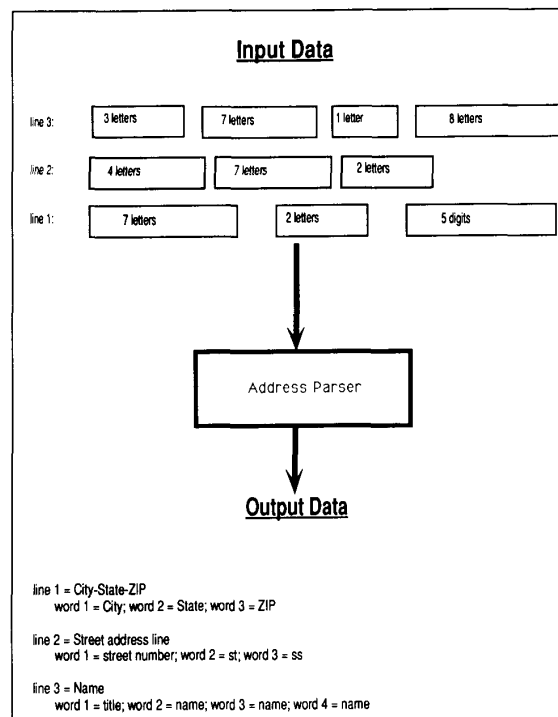
(a)



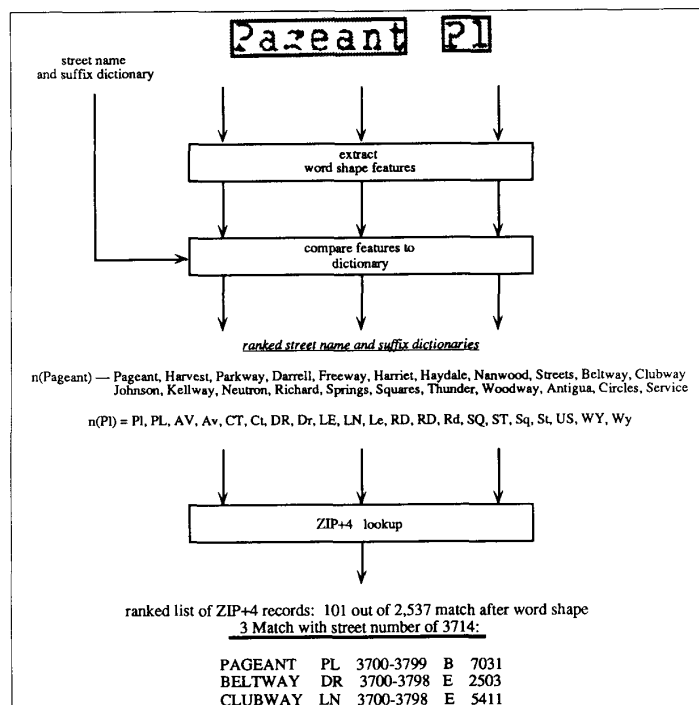
(b)



(c)



(d)



(e)

Fig. 14 Example of contextual analysis operations on a machine-printed address: (a) binary image rejected by postal OCR; (b) segmented lines; (c) segmented words; (d) parse results; and (e) word shape analysis results.

4 code specifies a small portion of a street, whereas a five-digit ZIP code specifies an entire city or a large portion of a city. The nine-digit ZIP + 4 code has been designed so that mail can be automatically encoded down to the level of individual carriers. Since the ZIP + 4 code is not present in a large proportion of the mail, it is necessary to read several lines of text in an address and look up the proper ZIP + 4 code in the U.S. Postal Service ZIP + 4 directory. A further desire of U.S. Postal Service mechanization efforts is to encode to carrier walk sequence. This means knowing the sequence in which carriers visit their customers and mail should be sorted in that order. It should be noted that the achievement of both goals depends on the ability to read and assign nine-digit ZIP + 4 codes.

The basic contextual address analysis (CAA) procedure is given in Fig. 12. Preprocessing consists of the following steps, performed in sequence (see Fig. 13): identify connected components, compute Hough transform (at several angles), remove underlines (possibly relabel components), and remove noise and correct for skew (additional Hough). The Hough transform [27] is used to determine evidence for lines of text at a given angle of skew. The primary Hough transform is done at several angles. The angles are finer near zero and get progressively coarser away from zero. Noise removal consists of removing small components. If the primary Hough transform indicated a high skew, the secondary Hough transform is performed at finer angles around the coarse skew indicated. After line segmentation, skew is verified using shadowing. Word segmentation is based on analyzing the vertical projection profile of the line. Tall and narrow peaks far from others are removed. Word segmentation produces alternative segmentation of a line based on confidences that a given number of words are present in a given line.

Parsing is the process of assigning type categories to each address field, e.g. street name and city name [28], [29]. Word recognition is performed by determining the word or words that can match the image of a word by utilizing the lexicon for its field. These words are termed the neighborhood of the word image. A neighborhood with only one word in it is equivalent to a complete recognition of the word image. The neighborhoods of the various words in an address are then used to access the ZIP + 4 file.

The assignment of the ZIP + 4 code involves matching the recognition results to one of nearly 30 million records in the ZIP + 4 file. The information that must be matched can be erroneous, incomplete, or missing altogether. We use a method that uses a constraint satisfaction approach [30]. It involves querying the ZIP + 4 file for the compatibility of each pair of recognized address fields and discounting those fields that are maximally incompatible. The raw ZIP + 4 file requires 4 gigabytes, which we compress to 400 megabytes (flat file + data structures), with an access time of 150 ms per query.

The operation of the individual steps are illustrated in Fig. 14. Part (a) shows the binary image of a machine-printed address that is rejected by a conventional OCR; parts (b) and (c) show the segmented lines and words; (d) shows the

result of a parser on the segmented words; and (e) shows the result of word recognition neighborhood finding. Thus, in this example the contextual analysis system is able to sort to five-digit accuracy and narrow down the four digit add-on to only three possibilities.

VII. DISCUSSION

As an area of computation research, machine reading continues to be of great economic and scientific importance. Among the basic literate skills of humans (reading, arithmetic, reasoning, and writing), a significant amount of success has been achieved in arithmetic (the *raison d'être* for early computers), in computational reasoning (e.g. expert systems), and in computational aids for writing (e.g. text processing, natural language synthesis). Computational reading, which spans the spectrum of signal processing to symbol processing, continues to be a challenging frontier.

In developing high-performance subsystems for a reading machine, a two-pass approach was taken. During the initial design, a computational theory was developed for each subsystem, and a prototype to demonstrate competence (in terms of ability to do the desired task without regard to speed) was developed. During the final design, the computational theory, algorithms, and representations were modified to achieve real-time performance on off-the-shelf and custom hardware while striving to maintain the original recognition rates.

Several design principles emerge from our design and implementation of a high-performance reading system. First, it is necessary to employ multiple algorithms based on diverse methodologies to achieve high recognition rates. This was observed in address block segmentation, handwritten digit recognition, handwritten word recognition, and degraded machine-printed word recognition. Second, among recognition methods, it is advantageous to employ recognition algorithms that operate at multiple scales—segmentation-based (or analytic) and non-segmentation-based (or holistic) approaches. Our solution to degraded machine-print recognition illustrates this. Third, contextual knowledge, in the form of lexicons generated by recognition of certain numeric and other fields, can lead to significant success in handwritten text interpretation. Fourth, large knowledge bases may have to be tightly coupled to the interpretation process.

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